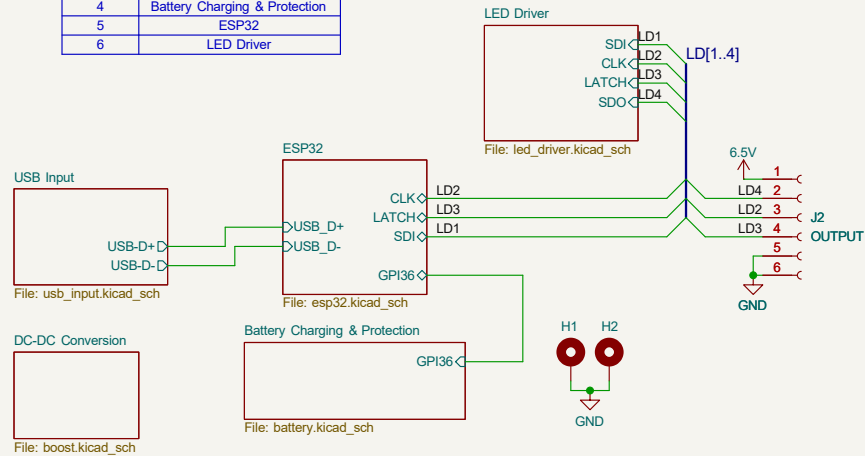
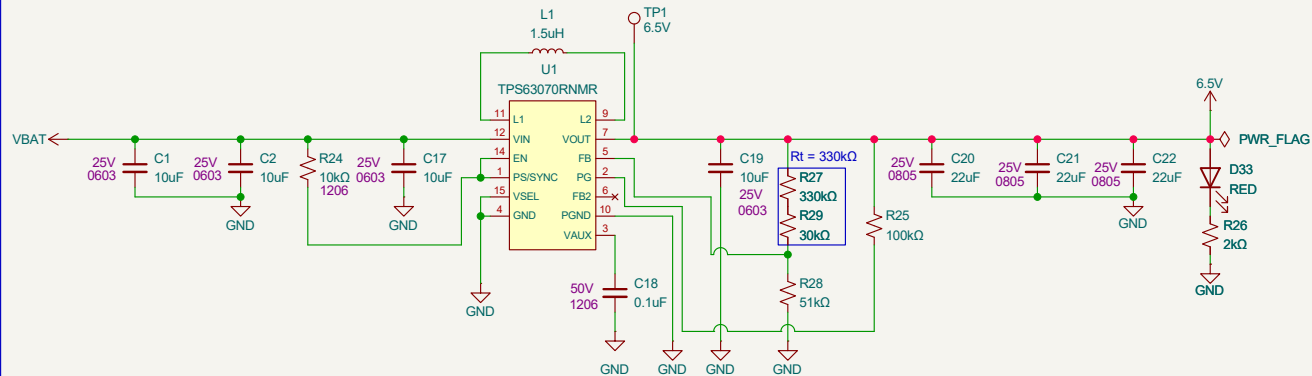


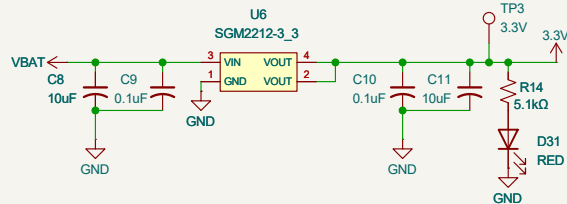
Page #	Content
1	Root
2	DC-DC Conversion
3	USB Input
4	Battery Charging & Protection
5	ESP32
6	LED Driver

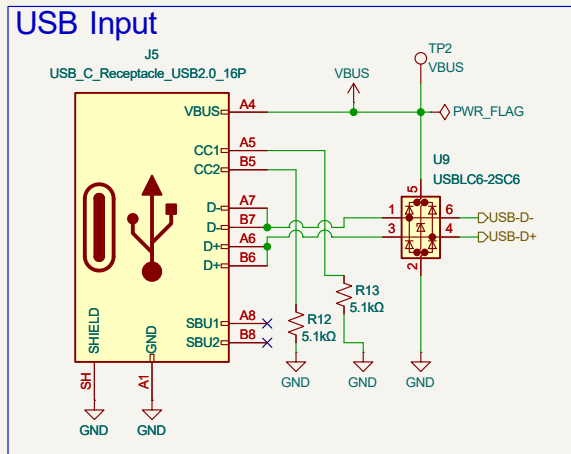


VBAT -> 6.5V

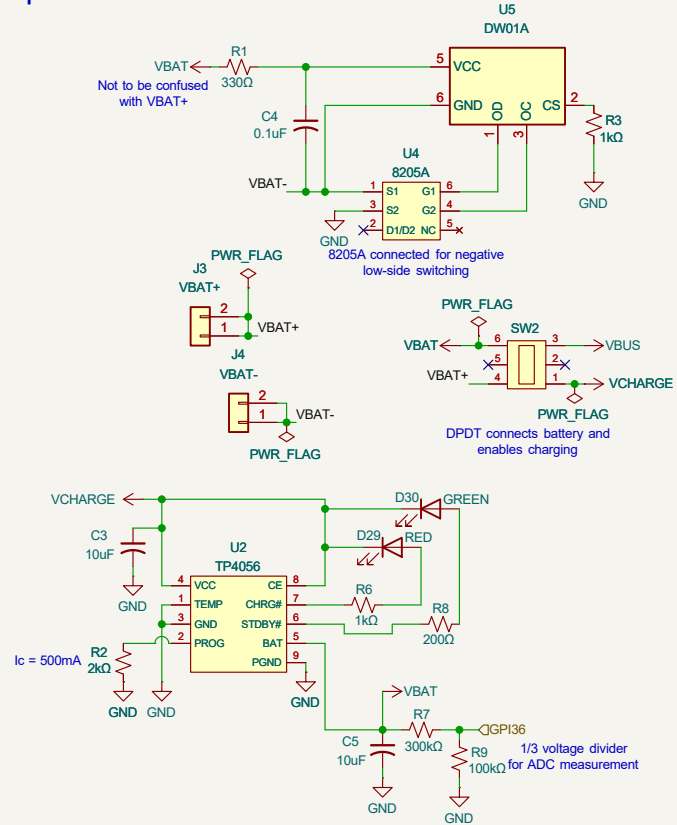


VBAT -> 3.3V





Battery charging & protection



BIRFLM

File: battery.kicad_sch Confidential & Proprietary

Size: User

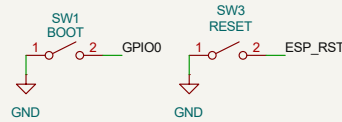
Date: 2026-10-04

Rev: v2.0

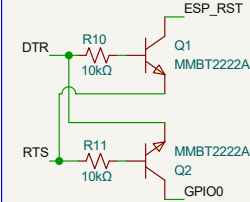
KiCad E.D.A. 10.0.2

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Boot & Reset

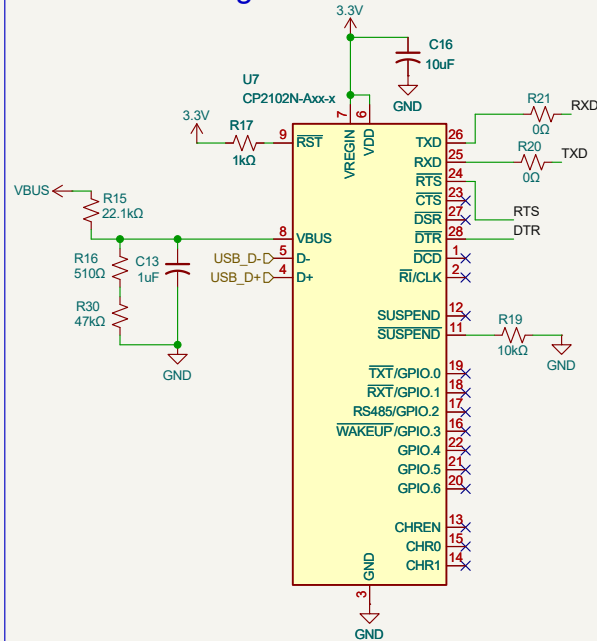


Auto-Reset Circuit

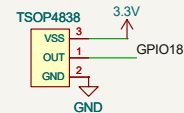
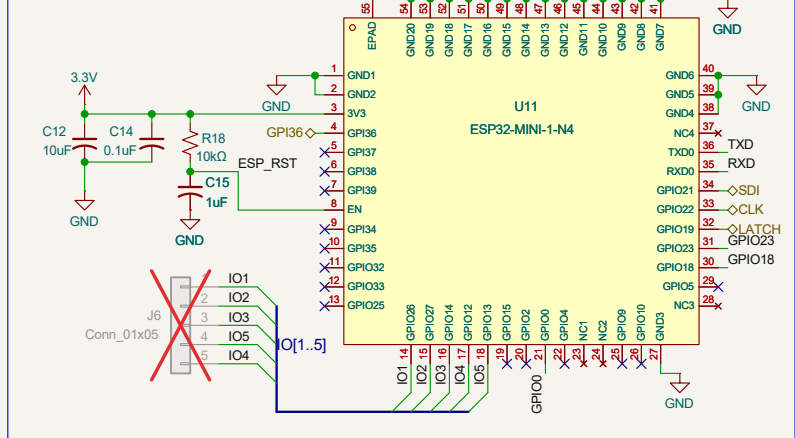


DTR	RTS	ESP_RST	IO0
1	1	1	1
0	0	1	1
1	0	0	1
0	1	1	0

USB-UART Bridge



ESP32-MINI-1 module



BIRFLM

File: esp32.kicad_sch Confidential & Proprietary

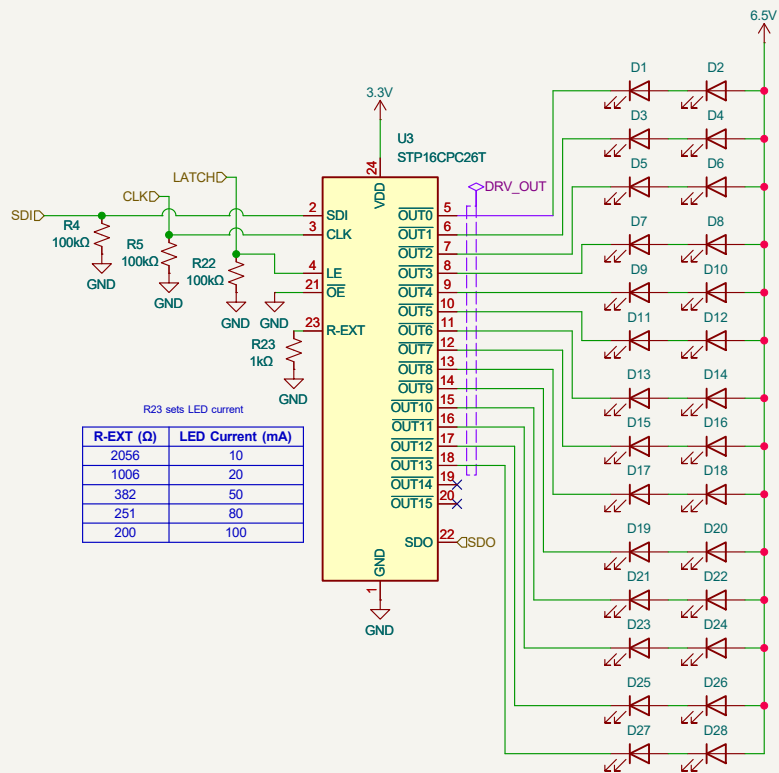
Size: User

Date: 2026-10-04

Rev: v2.0

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R-EXT (Ω)	LED Current (mA)
2056	10
1006	20
382	50
251	80
200	100



BIRFLM

File: led_driver.kicad_sch	Confidential & Proprietary	
Size: User	Date: 2026-10-04	Rev: v2.0
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